

Hybrid Drives: Get the Best of Both Worlds!



While hard disk drives were engaged in storing data on platters, the world moved on and solid-state drives made their appearance. SSDs, which are stacks of NAND flash memories, have their advantages as well as disadvantages. However, you can enjoy the best of both worlds by using hybrid drives, which incorporate both HDDs and SSDs.

Computer storage, specially the direct-attached type, began its journey around 1957 with IBM 325 systems, which were the first-of-a-kind 'platter' based disk drives available commercially. This led to people actually starting to consider storing data 'persistently' for their usage.

This revolution went through several milestones - advances in manufacturing processes and material sciences made the capacities of these 'platter' storage systems grow from megabytes to gigabytes in the 1990s, and to terabytes in 2007. Even today, these are the *de facto* first choice for personal computers, due to their potentially large capacity and high reliability.

A paradigm shift took place in this world of 'rotating platters' when the memory technology called 'NAND flash' emerged from Toshiba in 1984. In its initial form, NAND flash was intended to be used in handheld equipment for small capacity storage of microcode and data (EPROM like usage), but it soon became evident that we could stack up several such NAND flash chips together and still provide gigabytes of very fast drives in computers/laptops.

HDDs – Huge rotating storage

Taking a sneak peak into HDD design involves a deep dive into physics, kinetics and material sciences. Hard disk drives record the data by magnetising a thin film of ferromagnetic material. As depicted in Figures 1 and 2, a typical HDD design consists of a spindle that holds flat circular disks (or platters), which store the recorded data. The platters are made from a non-magnetic material (usually an aluminium alloy, glass or ceramic) and are coated with a thin layer of magnetic material (typically 10 to 20 nanometres thick). Data is recorded on these platters in concentric circles (called tracks, each containing several sectors of data) using a magnetic head or simply a head. Several such platters are put together and are made to spin at either 5,400 revolutions per minute or 7,200 revolutions per minute (and even higher in some HDDs) when the HDD is powered on. Data is binary encoded and read/written to the specific sector of the specific track of the specific platter using the head and actuator assembly driven by a motor.